

UGOLNYE KOPI, Russian Federation

WMO: 255630

Lat: 64.735N

Lon: 177.741E

Elev: 194

StdP: 14.59

Time Zone: 12.00 (E12)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.3	-32.1	-39.7	0.6	-35.1	-36.5	0.7	-31.9	55.8	16.9	51.7	18.9	14.0	0	1.015

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	69.6	59.2	65.5	56.8	62.0	55.1	60.0	67.2	57.6	63.7	55.8	60.9	11.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.5	68.5	63.4	54.5	63.7	60.9	52.8	59.7	58.9	26.5	67.4	25.0	63.9	23.8	61.2	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
48.2	39.9	32.8	DB	-38.9	76.2	3.8	3.1	-41.7	78.4	-43.9	80.2	-46.1	82.0	-48.9	84.2	
			WB	-38.5	63.9	4.2	2.3	-41.5	65.5	-44.0	66.9	-46.3	68.1	-49.4	69.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.6	-7.3	-5.4	-0.3	11.0	30.7	46.1	54.3	51.4	42.3	25.5	11.1	-1.3
	DBStd	25.18	16.53	17.27	14.43	11.73	7.85	7.48	5.75	5.52	6.58	9.73	14.22	17.47
	HDD50	10650	1776	1552	1561	1170	599	164	21	48	241	760	1167	1591
	HDD65	15827	2241	1972	2026	1620	1064	567	336	422	681	1225	1617	2056
	CDD50	301	0	0	0	0	0	47	153	91	10	0	0	0
	CDD65	4	0	0	0	0	0	0	3	1	0	0	0	0
	CDH74	19	0	0	0	0	0	4	13	2	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	14.0	15.0	16.0	15.1	12.9	12.4	13.0	13.1	12.1	12.1	13.5	17.0	16.0
Precipitation	PrecAvg	13.5	1.7	1.4	1.2	0.8	0.4	0.7	1.5	1.7	1.2	1.0	1.1	1.3
	PrecMax	24.3	4.6	5.8	4.1	3.3	1.3	2.4	4.1	3.4	3.6	3.1	3.1	5.0
	PrecMin	6.3	0.1	0.0	0.0	0.0	0.1	0.1	0.2	0.2	0.0	0.0	0.0	0.0
	PrecStd	3.8	1.0	1.3	1.1	0.7	0.3	0.4	0.9	0.8	0.8	0.7	0.8	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.3	32.3	33.2	35.5	53.2	73.3	75.5	71.6	61.1	46.8	37.8	33.9
		MCWB	31.8	31.4	31.8	32.8	44.7	58.1	62.4	60.6	53.4	44.8	36.7	32.6
	2%	DB	30.1	30.2	30.6	32.7	46.6	66.4	71.1	66.2	56.8	44.6	35.4	32.1
		MCWB	29.0	29.4	29.6	31.2	41.0	54.9	60.5	57.6	50.1	43.4	34.3	31.1
	5%	DB	25.0	26.9	28.0	31.0	42.6	62.0	66.8	62.3	53.5	40.7	33.2	29.9
		MCWB	24.0	25.8	26.9	29.7	38.2	52.6	58.2	55.8	48.9	38.8	32.4	28.8
	10%	DB	18.6	22.7	22.5	28.2	39.3	57.5	63.1	59.1	51.3	37.5	31.1	26.3
		MCWB	17.5	21.7	21.2	26.8	36.2	50.2	56.4	53.9	47.9	35.7	30.3	25.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.7	31.5	32.1	33.2	44.9	58.9	64.0	62.4	55.1	45.6	37.3	33.1
		MCDB	32.4	31.8	32.5	34.2	51.1	70.3	73.9	69.4	59.9	46.5	37.9	33.4
	2%	WB	28.7	29.2	29.7	31.4	41.1	55.8	61.2	58.5	51.5	43.0	34.2	31.1
		MCDB	29.8	29.9	30.7	32.4	46.0	65.1	69.2	64.3	54.6	44.5	34.8	31.8
	5%	WB	24.0	26.0	26.6	29.6	38.4	53.1	59.0	56.4	49.8	38.9	32.4	28.7
		MCDB	24.9	27.0	27.8	31.0	41.8	61.2	65.6	61.2	52.5	40.0	32.9	29.7
	10%	WB	17.4	21.7	21.2	26.6	36.3	50.6	57.0	54.6	48.2	35.8	30.2	25.2
		MCDB	18.5	22.7	22.2	28.0	38.9	57.1	62.3	58.5	50.8	37.2	31.1	26.3
Mean Daily Temperature Range	5% DB	MDBR	11.8	12.4	12.9	13.2	10.7	13.2	11.4	10.7	9.9	8.8	10.7	11.8
		MCDBR	14.9	13.3	12.2	10.4	15.5	20.5	17.2	15.4	13.0	8.2	10.3	11.5
		MCWBR	14.8	13.2	11.9	9.3	10.6	11.8	9.8	9.1	8.6	7.0	10.2	11.3
	5% WB	MCDBR	15.0	13.6	12.1	9.7	14.5	19.2	15.6	13.6	10.3	7.3	10.9	12.1
		MCWBR	15.0	13.6	12.0	9.0	10.2	11.4	9.7	8.9	7.5	6.7	11.0	12.1
Clear-Sky Solar Irradiance	taub	0.169	0.245	0.299	0.351	0.403	0.382	0.447	0.415	0.344	0.287	0.198	0.181	
	taud	1.929	2.133	2.068	1.952	1.973	2.293	2.129	2.196	2.370	2.254	1.948	1.927	
	Ebn at Noon	124	210	242	253	251	263	238	234	231	189	98	36	
	Edh at Noon	8	18	31	44	48	36	42	35	24	16	7	3	
All-Sky Solar Radiation	RadAvg	44	248	790	1472	1793	1788	1560	1176	721	286	72	14	
	RadStd	5	38	79	62	135	136	154	117	78	34	10	1	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+2.88	N/A	N/A	+2.79	+1.66	+1.80	N/A	N/A	N/A	N/A
Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page